TEXORA · LECTURE SHEET

Garments Testing

A garment fails where its fabric never could: at seams, buttons, zips and in the wash. This sheet covers testing the made-up article — seam strength and slippage, attachments and zips, dimensional change and appearance after washing, needle damage, care labelling, points of measure and the audit logic that turns measurements into accept or reject decisions.

TE-3201 · QUALITY EVALUATION OF FABRIC AND APPAREL · TOPIC 10 · LEVEL 3 TERM 2

1. Why garments are tested as made-up articles

Fabric tests certify a material; the customer wears a *garment*. Cutting releases tensions, sewing perforates yarns and joins fabrics of different stability, fusing bonds interlinings, and pressing sets new strains. The garment therefore has failure modes invisible in fabric testing: a seam that opens under a sound fabric, a button that pulls off through sound cloth, shrinkage that twists a leg seam although both fabrics passed their stability tests in isolation. Garment testing reproduces service on the finished article — or on production-representative sewn specimens cut from it — and is the final technical gate before shipment, alongside visual inspection and measurement audit. Specimens and whole garments are conditioned and handled in the standard atmosphere (65 % RH, 20 °C) wherever dimensions or strength are recorded.

2. Seam strength — ISO 13935

Seam strength is the maximum force a sewn seam withstands when pulled at right angles to the seam line. In ISO 13935 two routes are used: a strip method, in which a sewn specimen of defined width is pulled to rupture, and a grab method, in which only the central part of a wider specimen is gripped — closer to how a seam is loaded in wear. The record must state more than the peak force: *where* failure occurred decides the remedy. Failure modes are (a) fabric rupture away from the seam (the seam is stronger than the cloth — generally acceptable), (b) thread rupture (raise thread strength, stitch density or seam allowance design), and (c) yarn slippage / fabric rupture at the needle line (needle damage or excessive stitch tension — see Section 6). A high mean with thread ruptures at low individual values is a sewing-room problem, not a fabric problem, and the report that does not record failure mode sends the factory to fix the wrong thing.

WORKED EXAMPLE — SEAM STRENGTH: MEAN SPECIFICATION VS INDIVIDUAL VALUES

Five seam-strength readings (N): 185, 192, 178, 190, 180.

Mean = (185 + 192 + 178 + 190 + 180) / 5 = 925 / 5 = **185 N**.

Specification A — *mean* ≥ 180 N: 185 ≥ 180, the lot **passes**.

But one individual value, **178 N**, lies below 180 N. Under Specification B — *every individual* ≥ 180 N — the same data **fails**.

Discussion: neither specification is "wrong"; they protect different things. A mean specification controls average production quality and tolerates a low tail; an individual-minimum specification protects the wearer of the weakest garment. The contract must state which logic applies *before* testing. Where the mean passes but a low individual appears, good practice is to enlarge the sample, check that specimen's failure mode, and inspect the sewing station that produced it — not to average the problem away or to reject a lot on a rule that was never agreed.

3. Seam slippage (seam opening) — ISO 13936

A seam can open without anything breaking: the yarns of one fabric slide over the yarns of the other at the seam, opening a grin-through gap. This is **seam slippage**, the characteristic failure of smooth filament fabrics, loose weaves and low seam allowances. In ISO 13936 a sewn specimen is pulled at right angles to the seam under a defined procedure and the force required to produce a standard seam opening — conventionally **6 mm** — is recorded (fixed-opening method); the companion fixed-load method measures the opening produced by a standard load. A higher force at 6 mm means a more slip-resistant seam.

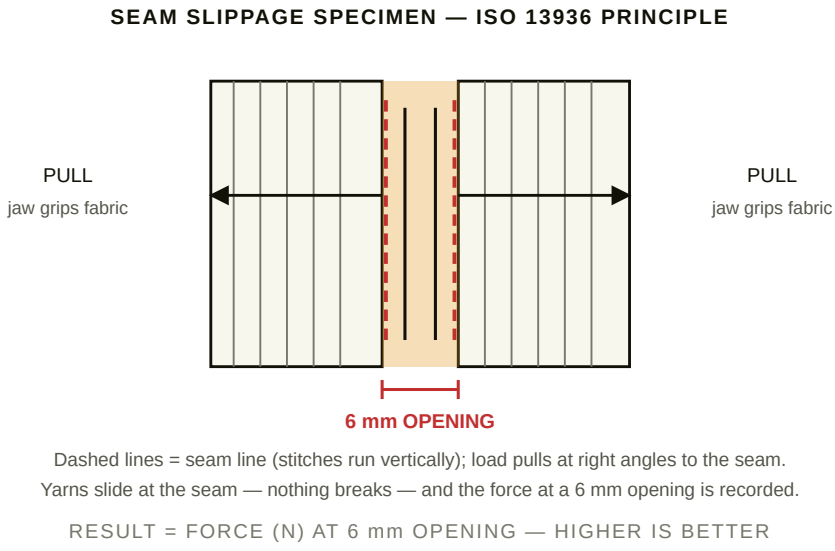


Figure 1 — Seam slippage test. The sewn seam lies across the pull direction; the specimen is extended until the seam has opened by 6 mm and the force at that opening is the result.

WORKED EXAMPLE — SLIPPAGE DECISION

A blouse fabric seam reaches the 6 mm opening at a load of **120 N**. The specification minimum is **110 N** at 6 mm.

120 ≥ 110 — **pass**, with a 10 N margin. Remedies if it had failed, in order of preference: increase seam allowance, raise stitch density, use a seam type that locks more yarns (for example a felled or safety-stitched seam), apply an anti-slip finish, or — last — change the fabric construction. Note that seam *strength* and seam *slippage* are independent: this seam might break at 220 N yet still grin open at 90 N, failing in service while "strong" on paper.

4. Attachments, zips and trims

4.1 Attachment strength of buttons and snaps

Buttons, press studs (snaps), rivets and eyelets are tested by pulling the component from the garment on a tensile tester or a dedicated pull tester and recording the detachment force and the failure mode — component breaks, thread breaks, or fabric tears out. The fabric-tear mode at low force is the safety-critical one, especially in childrenswear, where a detached small component is also a choking hazard and regulations impose minimum pull forces with a defined hold time. Production control uses go/no-go pull testers at the line; laboratory testing quantifies disputed lots. Findings feed back to stitch type, backing reinforcement and component quality — never only to "sew harder", which damages fabric and lowers the tear-out force.

4.2 Zipper (slide fastener) function

Zip tests, in outline, cover: crosswise strength of the closed chain, resistance to the slider pulling off the ends (top-stop and bottom-stop holding), reciprocating endurance — the slider is worked back and forth for hundreds of cycles and must still open and close smoothly — and lateral strength of the pin-and-box on open-end zips. A zip that passes strength but fails endurance will jam in the customer's first month; both families of test belong in a garment specification for zip-front products, together with a check that fabric cannot trap in the slider path by design.

5. Dimensional change and appearance after washing

Garments — not only fabric panels — are washed and dried by the care-label procedure for a stated number of cycles and then assessed as a set: (a) **dimensional change** at marked benchmarks, in length and width, reported as % shrinkage (−) or growth (+); (b) **smoothness appearance** — the washed garment is compared under standard lighting with photographic smoothness replicas and graded from 5 (smooth, as pressed) through 3.5 (the usual easy-care acceptance level) to 1 (heavily creased, unacceptable); colour change and staining are graded with the grey scales of Topic 7; (c) **seam and**

print appearance, fusing delamination or bubbling, and (d) **spirality (skew) in knitted garments**: after washing, a leg or side seam rotates around the body because of residual yarn torque, measured as the displacement of a seam or marker line from its intended position as a percentage of garment width. Spirality is a knitting-and-finishing property that only the washed garment reveals honestly; fabric-panel tests underestimate it because cutting and sewing constraints are absent. Acceptance commonly allows a few percent shrinkage with tighter limits on appearance-critical panels — the numbers belong in the purchase specification, not in a post-shipment argument.

6. Needle damage

Every stitch perforates the fabric. A correct needle parts the yarns; a wrong, blunt or burred needle cuts them, and the damage appears later as laddering in knits, seam-line rupture in wovens, or holes after washing when cut yarn ends withdraw. Needle damage is assessed by sewing a production-representative seam, then examining the seam line microscopically and by strength-testing the sewn specimen against an unsewn control — a strength loss at the needle line beyond the agreed allowance indicates cutting. A simple count method dyes or magnifies the seam and counts severed yarns per unit length. Prevention is a system: correct needle point and size for the fabric, scheduled needle changes (a needle is a consumable, not a permanent fitting), correct presser-foot pressure and feed so the fabric is not dragged against the needle, and re-checking after every needle-system change.

7. Care labelling — ISO 3758

Care labels translate laboratory wash behaviour into five standard symbols under ISO 3758, always in the same order: **washing** (tub — with temperature in °C and bars beneath for milder action), **bleaching** (triangle — plain for any bleach, crossed lines for oxygen bleach only, crossed-out for do not bleach), **drying** (square — circle inside for tumble drying with dots for heat level, lines for natural drying), **ironing** (iron — dots for temperature, crossed-out for do not iron) and **professional care** (circle — letters for solvent type, for wet-cleaning and dry-cleaning). A cross through any symbol prohibits that process. The label is a tested claim, not decoration: the recommended procedure must be one the garment survives with acceptable dimensional change, appearance and colour fastness — which is why the wash tests of Section 5 are run *by the label\'s own instructions*. A label that under-claims (“dry clean only” on a fully washable garment) is safe for the factory but misleads the consumer and, in many markets, breaches labelling rules that require the mildest adequate instruction set to be honest and complete.

8. Points of measure (POM) and tolerance practice

Fit is controlled by a **points-of-measure** list: for each style, named measurements with an exact measuring instruction — for a shirt, for example: chest (across, 2.5 cm below armhole), front length (highest shoulder point to hem), sleeve length (centre back neck over shoulder to cuff), collar size, cuff and hem widths. Every POM carries a tolerance, typically ±0.5 cm to ±1.5 cm depending on the dimension and product, measured on conditioned, relaxed garments on a flat table by the written method — the same POM measured by two different methods differs by more than its tolerance, so the method drawing is part of the specification. Auditors sample garments by an agreed sampling plan, measure every POM, and record actual values, not ticks: the data set, not an impression, drives the decision and the factory's corrective action. Systematic bias (all chests −0.6 cm) points to pattern or shrinkage-allowance error; wide scatter points to cutting or sewing control — the POM table distinguishes them at a glance.

9. Audit logic — mean vs individual-value specifications

Garment audit decisions combine the two specification logics met in Section 2. **Individual-value specifications** (every garment inside tolerance, every seam above a minimum force) protect each wearer and are used for safety and fit-critical items; sampling then follows an acceptance-quality-limit plan in which a stated number of defectives in the sample accepts or rejects the lot. **Mean specifications** (lot mean above a threshold) control average quality for properties where one soft result harms no one, such as average fabric mass. The costly disputes in the trade are almost never about measurement — they are about which logic the contract meant. A professional test report therefore states, for every property: the method, n, the individual values or their range, the mean, the specification *and its type*, and the verdict under that type. Testing a garment lot is, in the end, an exercise in agreed arithmetic performed on honestly made specimens.

Glossary

Seam strength — maximum force a seam withstands pulled at right angles to its line (ISO 13935). **Seam slippage** — yarn sliding at a seam, measured as force at a 6 mm opening (ISO 13936). **Attachment strength** — force to detach a button, snap or rivet. **Smoothness appearance** — washed-garment smoothness graded 5–1 against photographic replicas. **Spirality** — rotation of seams or wales in a knitted garment after washing, from yarn torque. **Needle damage** — yarns cut or weakened by the sewing needle. **Care label** — ISO 3758 symbol set for washing, bleaching, drying, ironing and professional care. **POM** — point of measure: a named garment dimension with a defined method and tolerance.

Key points

• Garments fail at joints and in the wash — test the made-up article, not only its fabric.
• Seam strength (ISO 13935): record the failure mode with the force; the mode decides the remedy.
• Seam slippage (ISO 13936) is independent of strength: result = force at 6 mm opening; 120 N passes a 110 N minimum.
• Mean specifications and individual-value specifications give different verdicts on identical data (185 N mean passes; a 178 N individual fails an every-value rule) — agree the type before testing.
• Care labels (ISO 3758) are tested claims; POM tables with written methods and tolerances turn fit into auditable data.

Practice

1. Seam strengths (N): 176, 188, 182, 195, 184. Calculate the mean. Decide the verdict under (a) a mean specification of ≥ 180 N and (b) an individual-minimum specification of 180 N.
2. A seam opens 6 mm at 95 N against a minimum of 110 N. State the verdict and list three sewing-room remedies in a sensible order.
3. Why can a seam record a high breaking strength yet still fail the customer in wear through slippage? Answer in terms of yarn movement.
4. A knitted T-shirt shows spirality after washing. Explain the mechanism and name two production stages where it could have been prevented or reduced.

Answer key

1. Mean = (176 + 188 + 182 + 195 + 184) / 5 = 925 / 5 = **185 N**. (a) Mean 185 ≥ 180 — **pass**. (b) The individual value 176 N is below 180 N — **fail** under an every-value rule. Same data, opposite verdicts: the specification type governs.
2. **Fail**: 95 N < 110 N at the standard 6 mm opening. Remedies in order: (1) increase seam allowance and/or stitch density so more yarns are locked; (2) change to a seam type that resists sliding better (felled or safety-stitched seam); (3) apply anti-slip finish or, if sewing changes are exhausted, tighten the fabric construction (higher thread density / less smooth yarn surface).
3. Strength measures the force to *break* something — thread or fabric. Slippage requires no breakage: warp and weft yarns simply slide over each other at the seam because friction and interlacing are low, opening a visible gap at loads far below the breaking load. A seam can therefore break at a high force in the laboratory test yet grin open in wear, where the load is modest but sustained.
4. Knitted loops are formed from yarn carrying residual twist torque. In the relaxed dry garment the torque is held by friction and finishing; washing plasticises the fibres, releases the torque, and each loop rotates slightly, accumulating as a visible skew of wales and side seams. Prevention: (1) at yarn/knitting stage — use balanced or plied yarns, set twist correctly, or knit with opposing-torque

feeders; (2) at finishing — heat-setting, resin or compacting treatments that lock the loop geometry, verified by a wash test on made-up garments before bulk production.
